

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

The de Broglie wavelength of a particle is given by $\lambda = h / \sqrt{2 m E}$, where λ is the de Broglie wavelength, h is the Planck's constant, m is the mass of the particle, and E is its kinetic energy. The kinetic energy of the electron is 1 eV, which needs to be converted into Joules. We know that $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. So, $E = 1.6 \times 10^{-19} \text{ J}$. The mass of an electron is approximately $9.11 \times 10^{-31} \text{ kg}$. Planck's constant h is approximately $6.626 \times 10^{-34} \text{ J s}$. Now, we can plug in the values to calculate the de Broglie wavelength: $\lambda = (6.626 \times 10^{-34} \text{ J s}) / \sqrt{2 \times 9.11 \times 10^{-31} \text{ kg} \times 1.6 \times 10^{-19} \text{ J}}$. Simplifying the expression gives the de Broglie wavelength. Converting the answer to centimeters and comparing with the given options will give us the correct answer.

</think><answer> G. 1.23×10^{-7} cm